

COMPUTER ARCHITECTURE

PRESENTATION



TOPIC

PROCESSOR STRUCTURE AND FUNCTION



WHAT IS PROCESSOR (CPU)?

- ▶ The Central Processing Unit (CPU) is the brain of the computer. It performs all calculations, decision-making, and controls system operations.

MAIN RESPONSIBILITIES:

- 01 Execute instructions
- 02 Process data
- 03 Control other components





BASIC STRUCTURE OF PROCESSOR

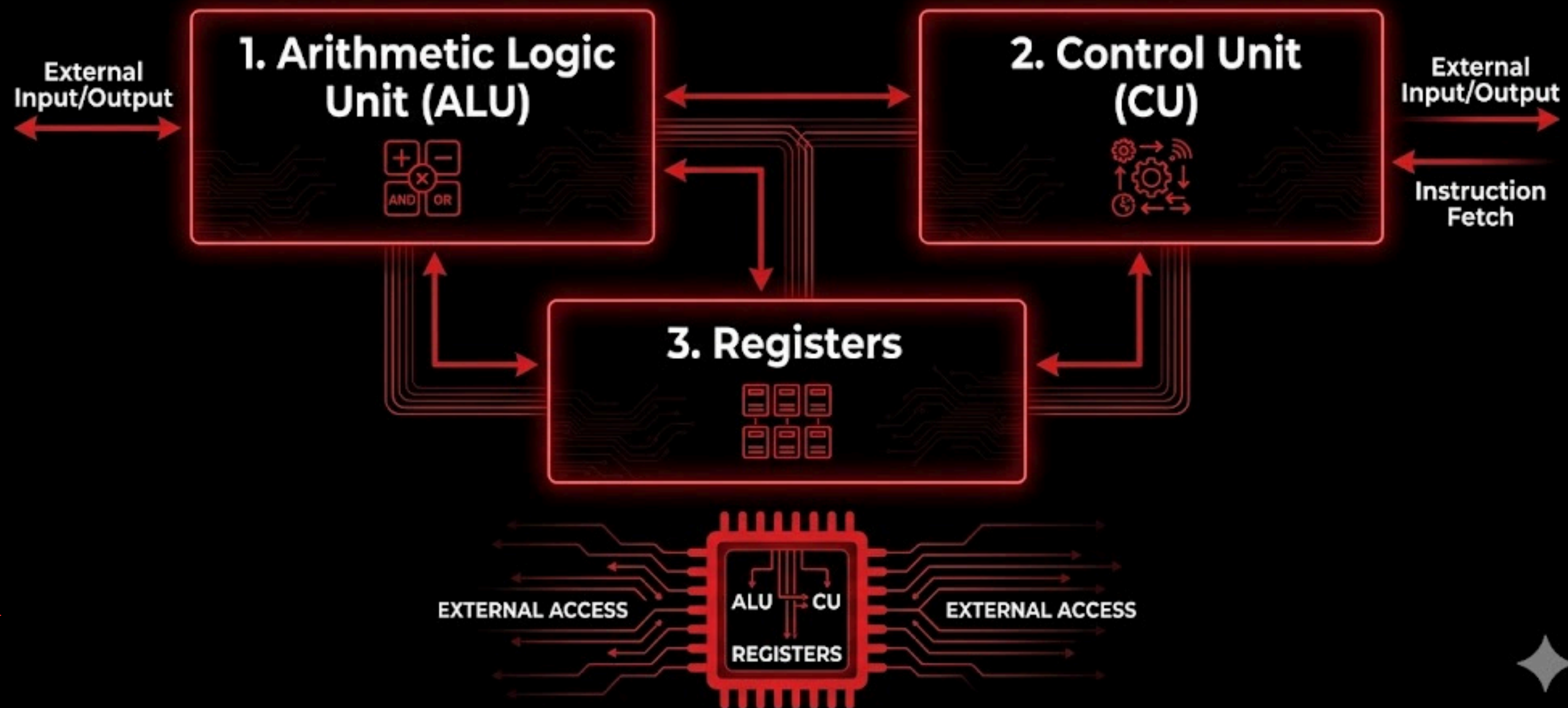
→ A PROCESSOR IS MAINLY DIVIDED INTO THREE CORE COMPONENTS:

1. Arithmetic Logic Unit (ALU)
 2. Control Unit (CU)
 3. Registers
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A PROCESSOR IS MAINLY DIVIDED INTO THREE CORE COMPONENTS:

PROCESSOR CORE ARCHITECTURE





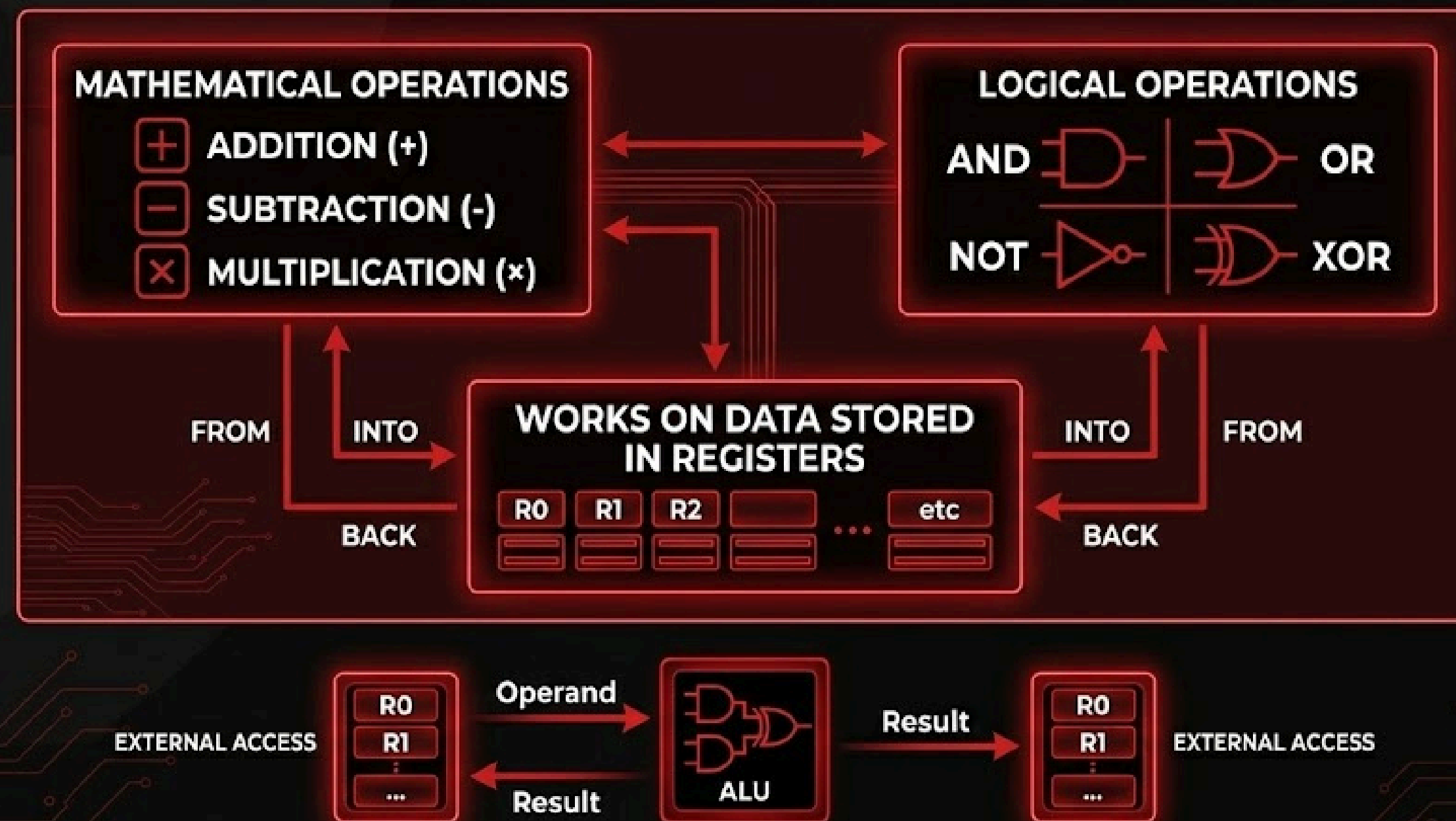
BASIC STRUCTURE OF PROCESSOR

- 1. Arithmetic Logic Unit (ALU)
 - Performs mathematical operations:
 - Addition, subtraction, multiplication
 - Performs logical operations:
 - AND, OR, NOT, XOR
 - Works on data stored in registers



ARITHMETIC LOGIC UNIT (ALU) - DETAILED OPERATION

ARITHMETIC LOGIC UNIT (ALU)





BASIC STRUCTURE OF PROCESSOR

→ 2. Control Unit (CU)

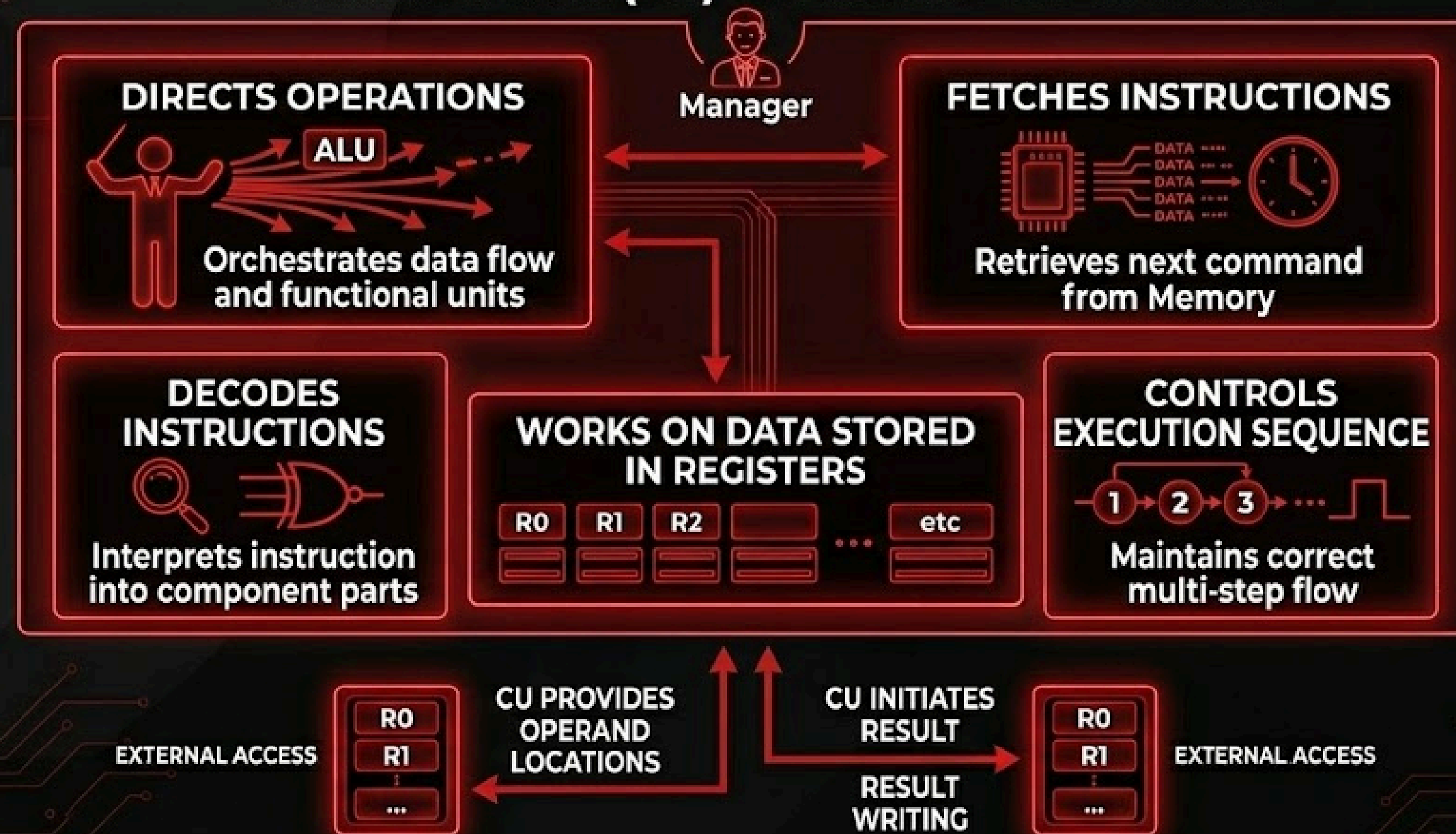
- Directs all operations of the processor
- Fetches instructions from memory
- Decodes instructions
- Controls execution sequence

Think of CU as a manager controlling all activities.



CONTROL UNIT (CU) - THE PROCESSOR MANAGER

CONTROL UNIT (CU) - DETAILED OPERATION





BASIC STRUCTURE OF PROCESSOR

→ 3. Registers

- Small, fast storage inside CPU
- Store temporary data and instructions

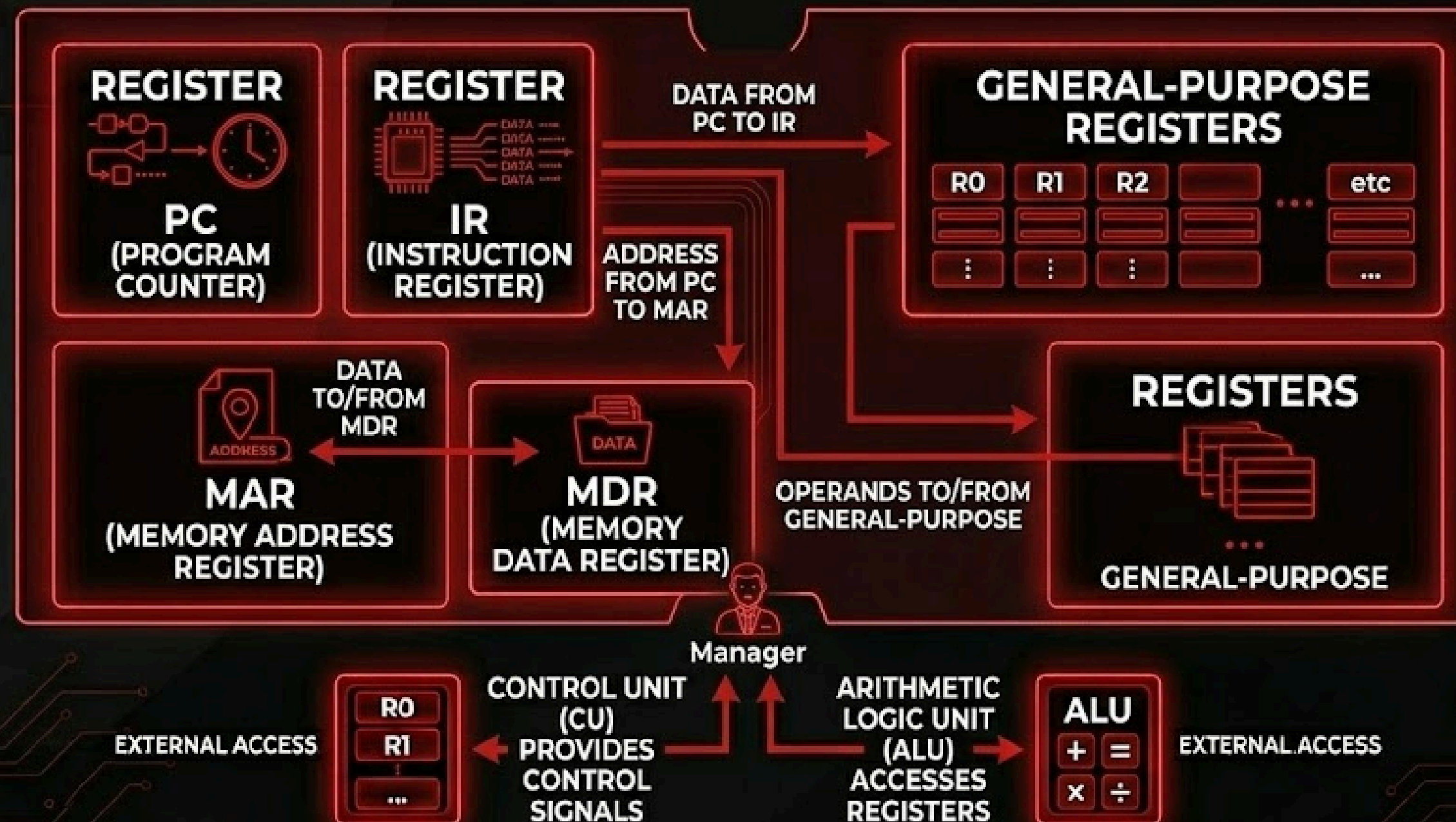
→ Types of registers:

- Program Counter (PC)
- Instruction Register (IR)
- Memory Address Register (MAR)
- Memory Data Register (MDR)
- General-purpose registers



REGISTERS: SMALL, FAST STORAGE INSIDE THE CPU

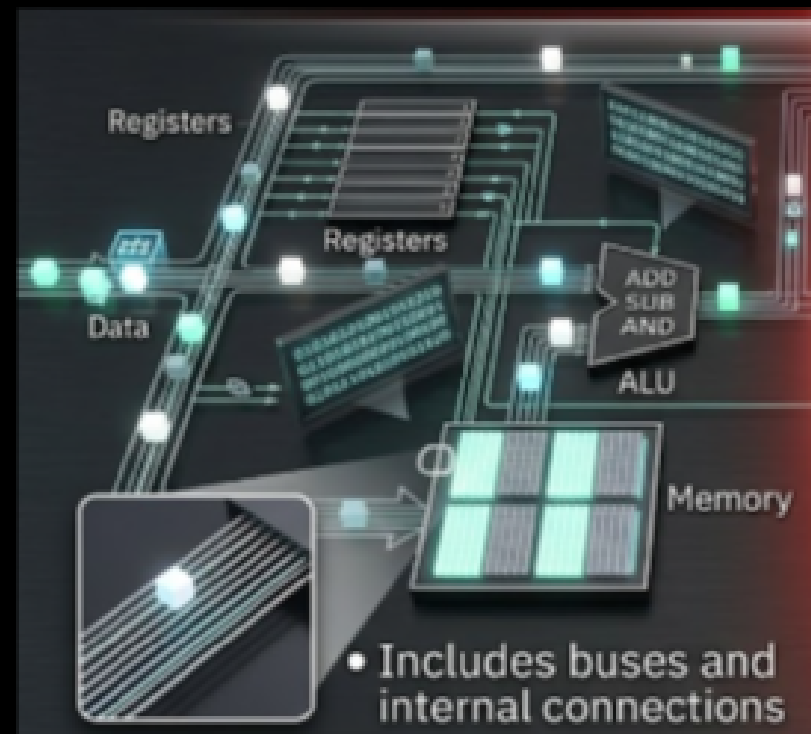
TYPES OF CPU REGISTERS & THEIR ROLES



PROCESSOR FUNCTIONAL UNITS

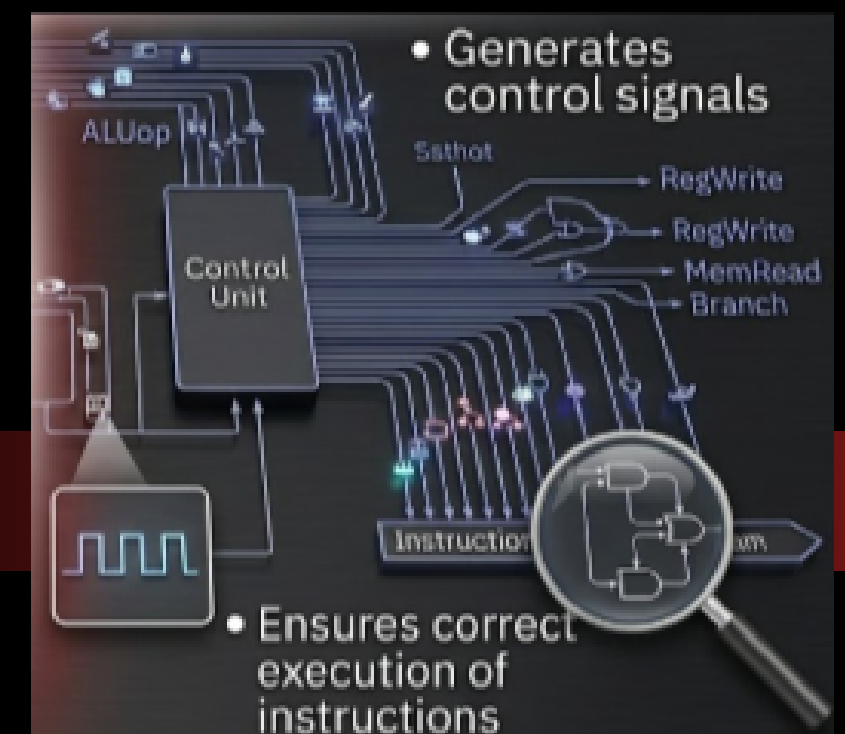
▶ Data Path

- Transfers data between registers, ALU, and memory
- Includes buses and internal connections

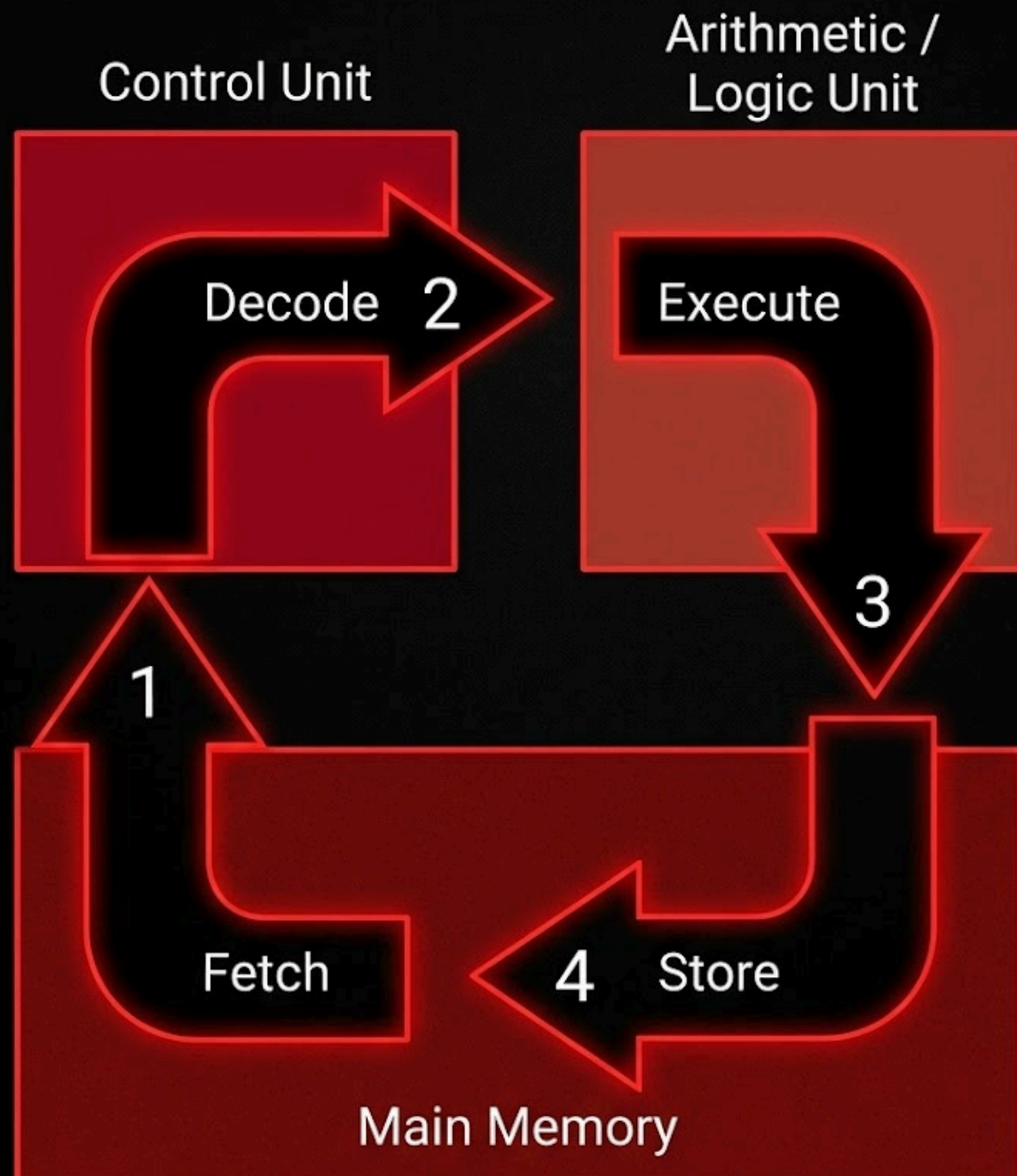


▶ Control Path

- Generates control signals
- Ensures correct execution of instructions



Instruction Cycle (Fetch–Decode–Execute)

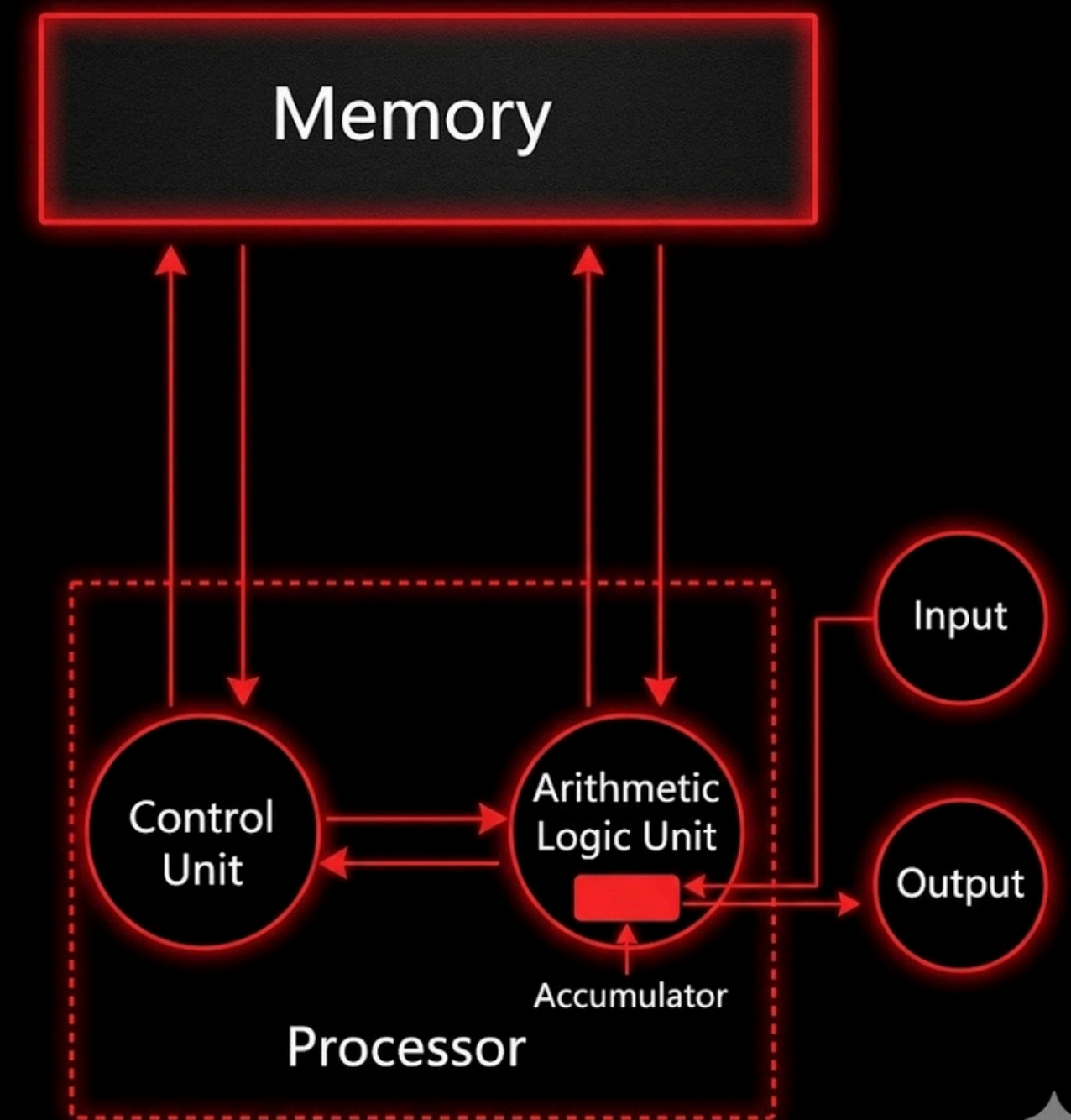


- 01** Fetch
 - Instruction is fetched from memory
 - Address stored in Program Counter (PC)
- 02** Decode
 - Instruction is interpreted by Control Unit
- 03** Execute
 - ALU performs operation
 - Result stored in registers or memory

This cycle repeats continuously.

TYPES OF PROCESSOR ARCHITECTURE

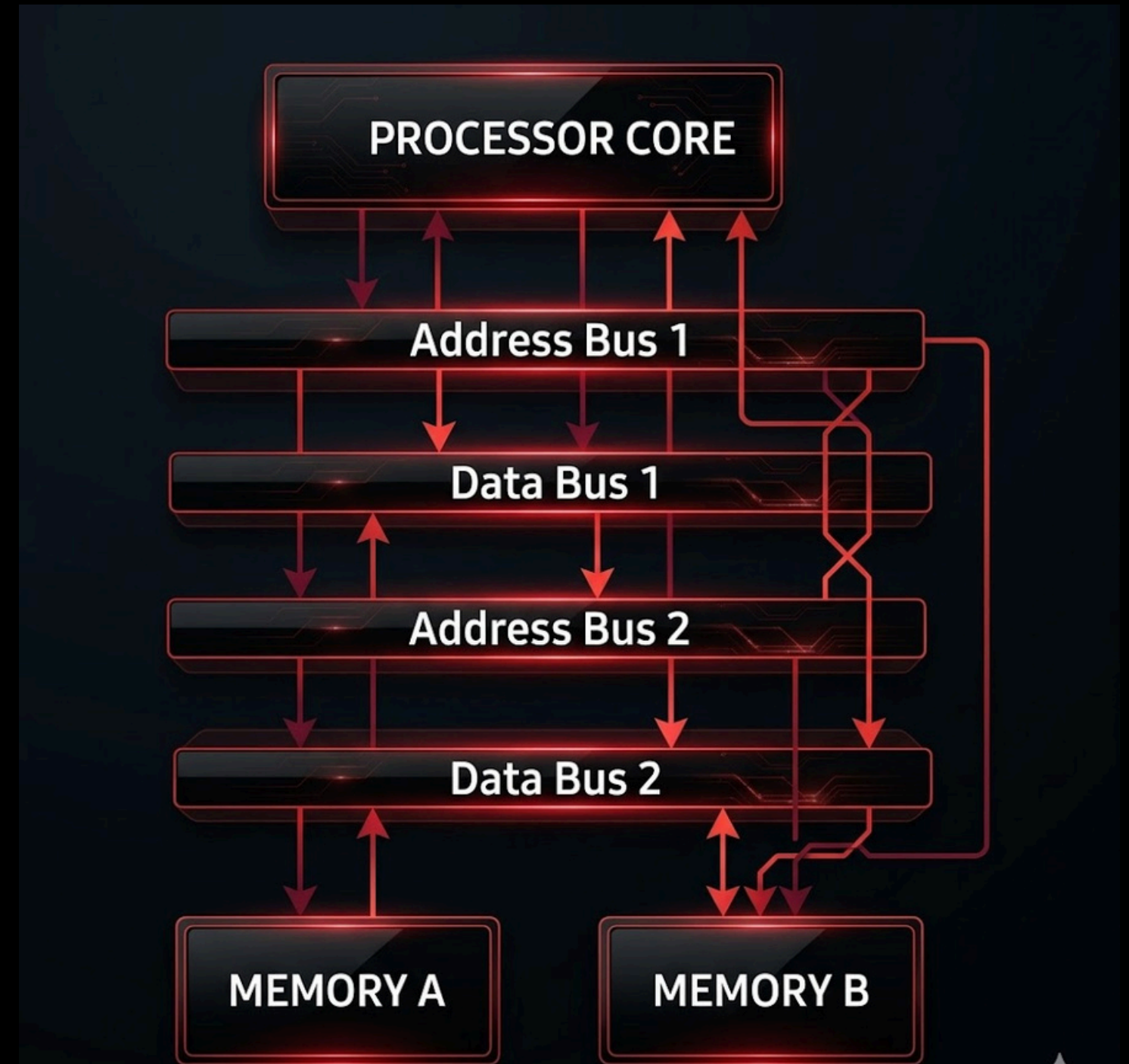
- ▶ 1. Von Neumann Architecture
 - Single memory for data and instructions
 - Simpler but slower (bottleneck issue)



TYPES OF PROCESSOR ARCHITECTURE

▶ 2. Harvard Architecture

- Separate memory for data and instructions
- Faster and more efficient



PROCESSOR PERFORMANCE FACTORS

▶ Key factors:

- Clock speed (GHz)
- Number of cores
- Cache memory
- Instruction set design



TYPES OF PROCESSORS

1. Single-Core Processor

A single-core processor has only one core (processing unit), so it can handle one task at a time.

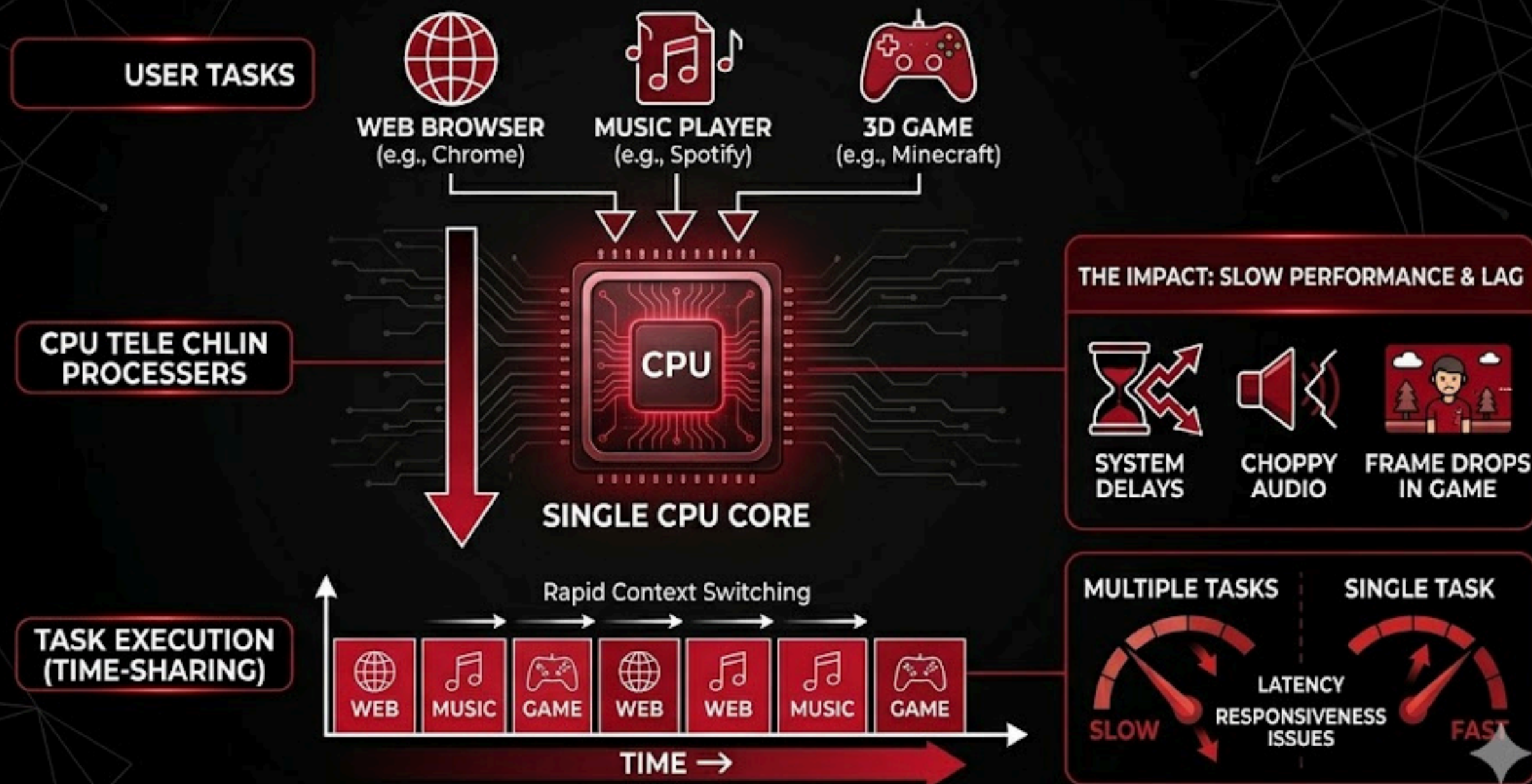
- Executes instructions sequentially
- Uses time-sharing to switch between tasks

Example:

- Old computers
- Early versions of Windows XP

TYPES OF PROCESSORS

• SINGLE-CORE PROCESSOR LIMITATIONS • - HANDLING MULTIPLE TASKS ON ONE CORE



TYPES OF PROCESSORS

2.A MULTI-CORE PROCESSOR

It has multiple cores, and each core can execute tasks simultaneously.

Types:

- Dual-core (2 cores)
- Quad-core (4 cores)
- Octa-core (8 cores)

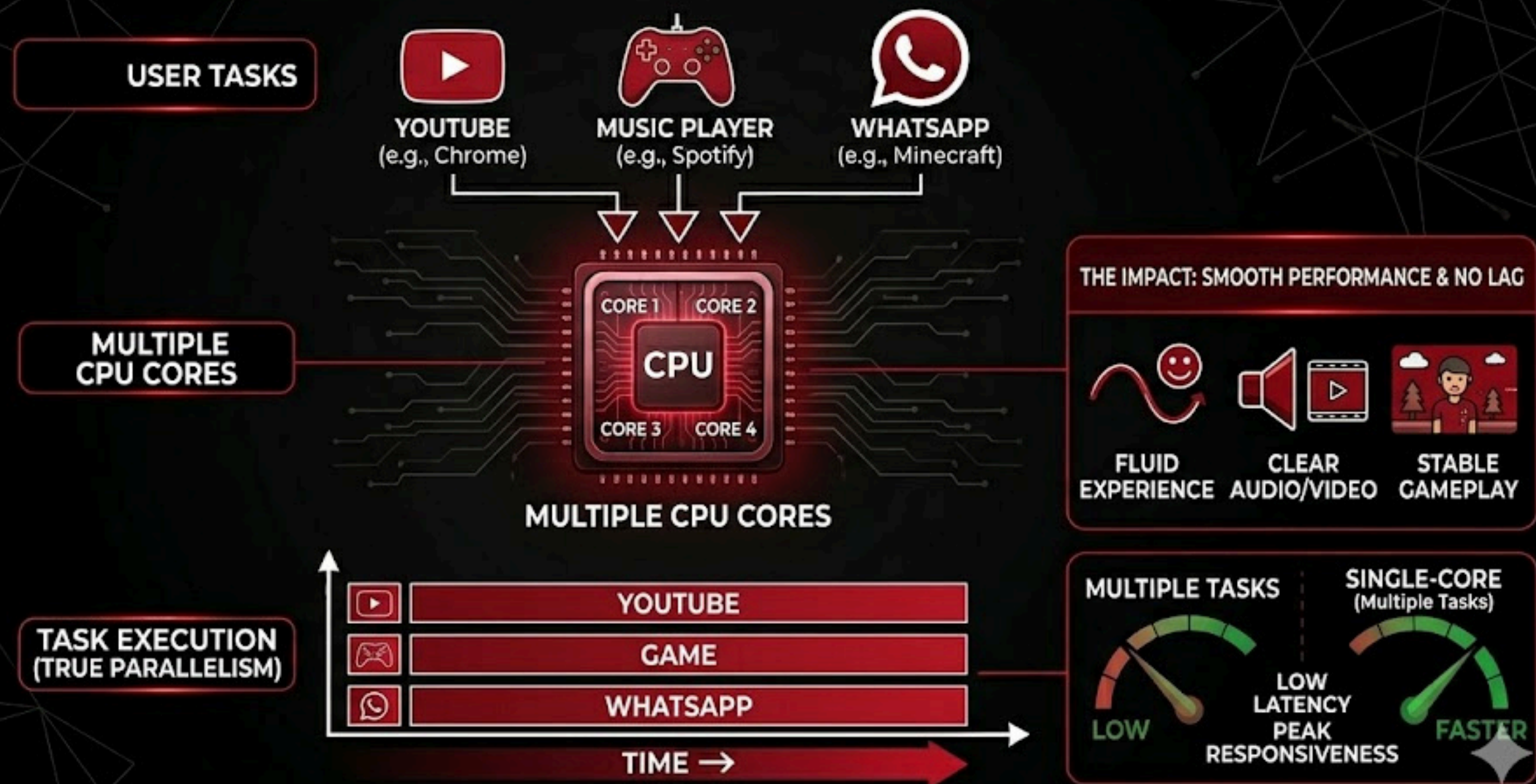
Example:

Modern laptops and smartphones like:

- Gaming PCs
- Android phones

TYPES OF PROCESSORS

• MULTI-CORE PROCESSOR ADVANTAGES: • - TRUE SIMULTANEOUS TASKING



TYPES OF PROCESSORS

3. EMBEDDED PROCESSORS

An embedded processor is designed for a specific task, not general-purpose computing.

- Mobile phones
- Cars (engine control system)
- Washing machines
- Smart devices (IoT)

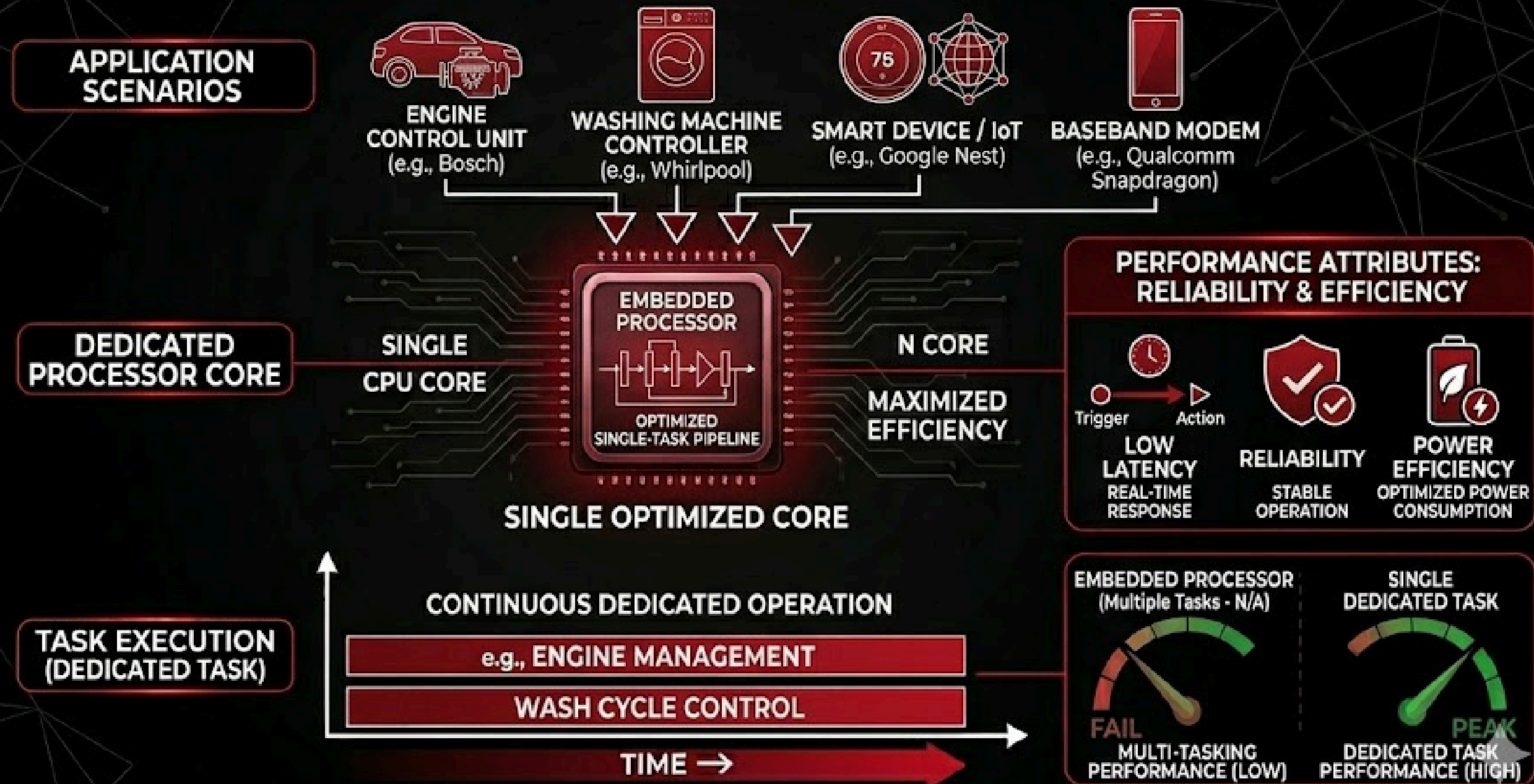
Example:

- Microwave oven timer
- Smart TV system
- Fingerprint sensor

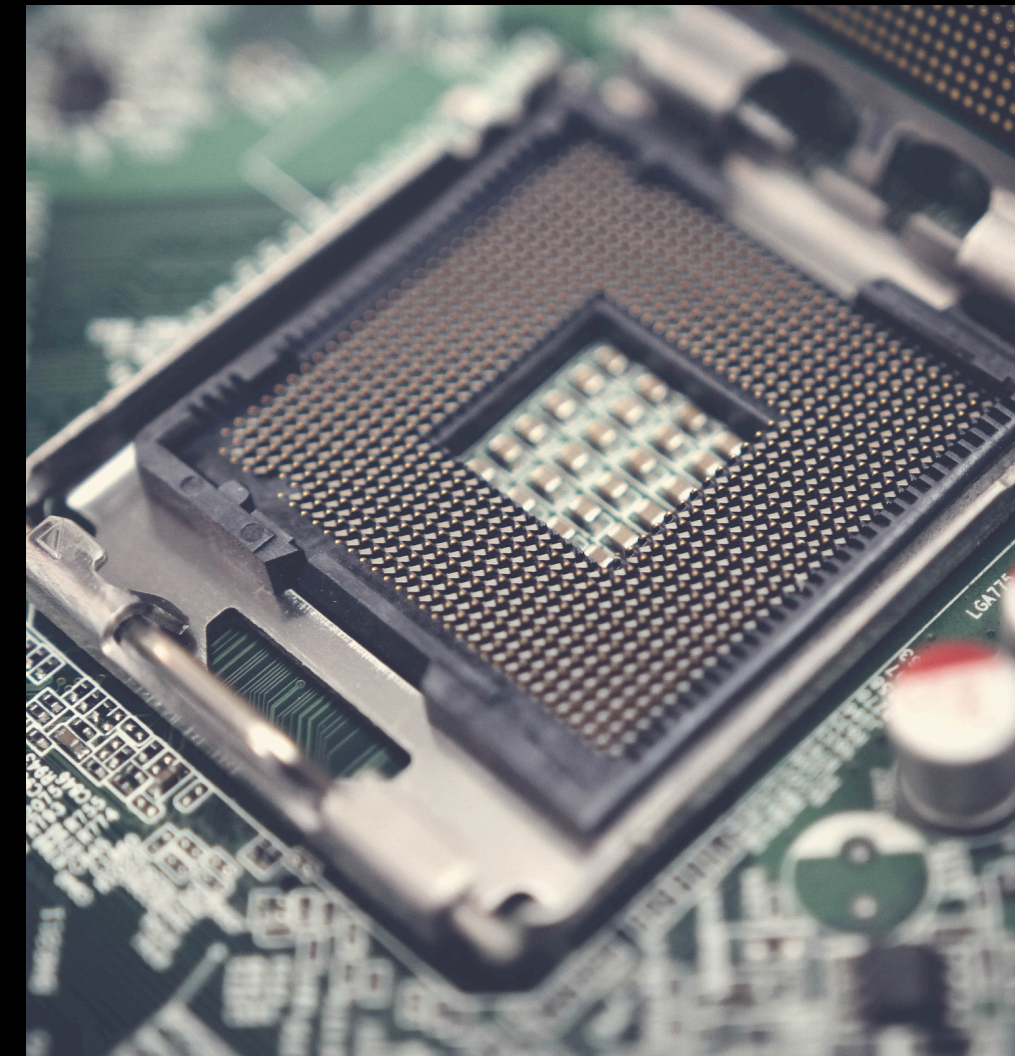
It performs one dedicated job

TYPES OF PROCESSORS

EMBEDDED PROCESSOR ADVANTAGES: DEDICATED, SPECIFIC-TASK PROCESSING



FUNCTIONS OF PROCESSOR



01 Execute instructions

03 Control data flow &
Handle interrupts

02 Perform arithmetic &
logic operations

04 Manage input/output
operations

MODERN PROCESSOR FEATURES

01

Pipelining

02

Parallel processing

03

Multithreading

04

Cache
optimization

CONCLUSION

- Processor is the central unit of computation
- Combines ALU, CU, and Registers
- Works using Fetch–Decode–Execute cycle
- Performance depends on architecture and design

THANK YOU

ANY QUESTION?